

Spandan Dhru

443-768-3956 | sdhru29@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

EXPERIENCE

M&T Bank

July 2025 – Present

Software Engineer

Buffalo, NY

- Designed a generic Java Spring Boot batch framework consolidating **16+ request/response classes** across **6 payment rails**, and exposed it through versioned **REST API** endpoints.
- Diagnosed a batch processing defect that silently skipped scheduled transactions, restoring reliable processing in a credit card microservice handling **350K+ monthly** transactions.
- Built an AI coding agent, adopted by engineers **across 10 repositories**, that pulls SonarQube findings, validates fixes locally, and opens GitLab merge requests automatically.
- Migrated 3 payment services from AWS S3 to Azure Blob Storage with **zero downtime** by designing a dual-provider storage abstraction, part of a 20+ service team-wide migration.
- Standardized incident triage with **10 bounded SQL Server diagnostic views** across 4 database domains, replacing ad-hoc queries on production tables.
- Cut manual reporting from **4 hours to under 10 minutes** for 10 teams by building a reusable Python FastAPI service delivered as a GitLab CI/CD component.

Immuta

May 2024 – December 2024

Software Engineer Intern

College Park, MD

- Shipped full-stack policy management features across Angular, TypeScript, PostgreSQL, and Prisma for the primary interface managing **10K+ data access policies**, increasing policy author satisfaction by **15%**.
- Extended Node.js REST API endpoints for **domain creation** in the policy engine, adding PostgreSQL-specific support, server-side pagination, and feature-flagged rollout.
- Resolved a high-priority tag search defect affecting **enterprise customers** by refactoring the tag filter API for exact-match lookups while preserving fuzzy search elsewhere.

University of Maryland

Jan. 2022 – Aug. 2023

Undergraduate Researcher

College Park, MD

- Built a deep neural network classifying Higgs boson decays across 14K+ simulated LHC samples, outperforming a shallow baseline on AUC and benchmarking against a Qiskit quantum classifier.

PROJECTS

Yorm: Real-Time Tabletop RPG Server | Go, TypeScript, PostgreSQL

Sept. 2026 – Present

- Built an authoritative real-time game server in Go with per-session goroutines and PostgreSQL event sourcing, restoring **5,000-event sessions in ~5 ms** from snapshots.
- Cut p99 latency from **11.8 s to 38 ms at 3,000 simulated concurrent WebSocket clients** by batching commits across sessions, using single-statement appends, and sizing the connection pool from metrics.
- Guaranteed **zero lost or duplicated actions** across server crashes through idempotent commands and reconnect catch-up, verified by kill -9 chaos tests.

Savant: Chess Coaching Agent | Python, Stockfish, OpenAI API, python-chess, SQLite

July 2026 – Present

- Engineered a two-tier analysis pipeline routing per-position evaluation to Stockfish and reserving LLM calls for critical mistakes, cutting LLM calls by **93%** versus LLM-only analysis.
- Built incremental sync from the Chess.com and Lichess APIs into a unified SQLite schema, powering a RAG coach that answers natural-language questions across **5K+** games.

SKILLS

Languages: Java, Python, Go, TypeScript, JavaScript, SQL

Frameworks & Libraries: Spring Boot, Spring Data JPA, FastAPI, Angular, React, Node.js, JUnit, Mockito

Databases: Microsoft SQL Server, PostgreSQL, SQLite

Cloud & DevOps: AWS, Azure, OpenShift, Kubernetes, Docker, GitLab CI/CD, Maven

AI & Developer Tools: Claude Code, Codex, OpenCode, GitLab Duo, Git, Postman

EDUCATION

University of Maryland, College Park

College Park, MD

B.S. in Computer Science

Aug. 2021 – Dec. 2024